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Paleomagnetic constraint on the formation of the Eastern Himalayan Syntaxis: A new late Eocene result from the Mangkang area of the eastern Tibetan Plateau



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ABSTRACT

The Eastern Himalayan Syntaxis (EHS) is a major structural and geomorphological unit in Eastern Tibet; however, its mechanism of formation and the extent of its northward effect of compression remain unclear. To address these questions, we conducted a paleomagnetic and geochronologic study of the late Eocene trachyte intrusive mass in the Mangkang area of Tibet. The U-Pb SHRIMP II age estimate of the trachyte intrusive mass is ~40 Ma, and the estimated mean paleomagnetic direction we obtained is D=27.0°, I=36.8°, k=42.9, α_{95}=1.3°, and n=282 specimens. These results indicate that the Mangkang area was located at a paleolatitude of 20.5 ± 0.9° N during the late Eocene. A comparison of paleolatitudes of the Qiangtang block with those inferred from the apparent polar wander paths (APWP) of India and Eurasia suggests that the Eastern Himalayan Syntaxis was formed after 35 Ma when Eurasia was penetrated by ~600 km by northward-moving India. This is consistent with geologic evidence of intra-continental shortening in the foreland of the Eastern Himalayan syntaxis. The northward counterclockwise penetration of the corner of India also resulted in the large right-lateral strike-slip system of the Sagaing fault (SGF), with ~400-600 km of displacement between the EHS and the Shan-Thai block. Thus, the northward wedging of the EHS indicates minimal southward displacement of the Shan-Thai block. Our results link the tectonic activity of the EHS to the lateral extrusion of the southeastern margin of the Tibetan Plateau, providing a new perspective on the evolution of this region since the late Eocene.

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1. Introduction

The continental collision between the Indian and Eurasian plates in the early Eocene resulted in strong continental shortening that formed the Tibetan Plateau, which has the highest average elevation of any plateau on Earth. This process formed a series of large strike-slip fault zones in the eastern part of the plateau. A common view is that since the India-Eurasia plate ultimate collision, the Indian plate experienced continuous counterclockwise rotation during 65–45 Ma, which resulted in its northeast corner penetrating the Eurasian continent to form an arc-shaped structure called the Eastern Himalayan Syntaxis (EHS) (Treloar and Coward, 1991; Harrison et al., 1992; Jain and Manickavasagam, 1993; Guillot et al., 1999; Ding and Zhong, 2013; Yuan et al., 2021). The Tibetan Plateau consists of a collage of elongated blocks, in which the Lhasa and the Qiangtang terranes extend from east to west before being rotated 90° to the north-south orientation after passing through the EHS, and the response to the India-Asia collision changes from north-south crustal shortening to southeastward crustal material escape (Armijo et al., 1986; Tapponnier et al., 1982; Royden et al., 1997; Wang and Burchfiel, 2000; Jessup et al., 2008). The displacement of major strike-slip faults and large-scale clockwise rotation of the southeastern edge of the Tibetan Plateau occurred, which was caused by the extrusion of the continental blocks (Tapponnier et al., 1982; Tong et al., 2021).

Recent studies indicate that the Indian plate collided with the West Burma block after the late Eocene, after which they com-

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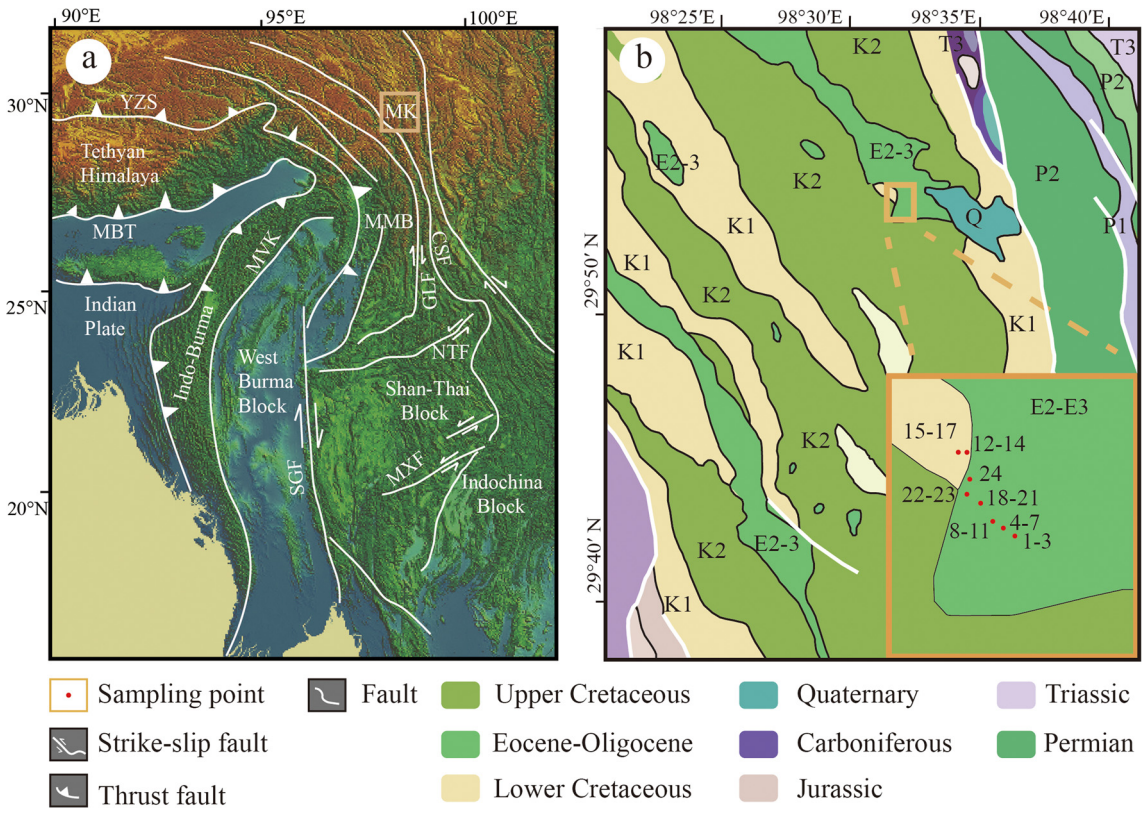


Fig. 1. (a) Schematic tectonic geologic map of Southeast Asia. (b) Schematic tectonic geologic map of the Mangkang area and distributions of sampling sites of the present work. GLF, Gaoligong strike-slip fault. MVK, Mount Victoria-Kawlun belt; NTF, Nantinghe strikeslip fault. SGF, Sagaing strike-slip fault.CSF, Chongshan strike-slip fault. MMB, Mogok metamorphic belt. MXF, Mengxing strike-slip fault. YZS, Yarlung Zangbo suture zone. MBT, Main boundary thrust. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

menced north-northeastward movement into the eastern part of the Himalayan tectonic zone, initially forming the East Himalayan Syntaxis. This promoted the large-scale rotation of the blocks on the southeastern edge of the Tibetan Plateau and caused the boundary strike-slip fault zones to commence their main phase of activity (Westerweel et al., 2020). Tong et al. (2021) proposed that the northward indentation of the Indian plate and West Burma block into the Eastern Himalayan orogenic belt formed the EHS after ~28–26 Ma. Therefore, knowledge of the tectonic evolution of the EHS is important for understanding the tectonic evolution of the eastern and southeastern margins of the Tibetan Plateau and the lateral escape of crustal materials.

Mangkang is located at the front of the EHS, where late Eocene–Oligocene volcanic rocks were extruded. Paleomagnetic methods can be used to reconstruct the tectonic changes of the EHS foreland block when it was indented northward by the northeast corner of India. To this end, we conducted a paleomagnetic study in the Mangkang area on the northern margin of the EHS in the southeast part of the Qiangtang terrane in the Tibetan Plateau. Our objectives were to estimate the northward displacement of the Eastern Himalayan Syntaxis since the late Eocene and to elucidate the process of formation of the EHS and the influence of wedging on the tectonic evolution of the strike-slip fault zone on the southeastern margin of the Tibetan Plateau.

2. Geologic background and sampling

The interior of the eastern Tibetan Plateau consists of the Songpan-Ganzi terrane, the Qiangtang terrane, and the Lhasa terrane. The Qiangtang terrane extends roughly west-east, passing the EHS, and it then turns 90° in a north-south direction and extends to the southeast, gradually being replaced by the Shan-Thai Block. The EHS divides the Tibetan Plateau into two distinct tectonic domains: north-south crustal shortening initially, followed by east-west extension in the east (Armijo et al., 1986; Jessup et al., 2008); and southeastward displacement of crustal material in the eastern and southeastern Tibetan Plateau (Tapponnier et al., 1982; Royden et al., 1997; Wang and Burchfiel., 2000). Mangkang is located in the southeastern part of the Qiangtang terrane (Fig. 1), where two tectonic domains are connected and directly affected by the intrusion of the EHS.

The igneous activity was concentrated during the Eocene–early Oligocene in the Qiangtang terrane, characterized by the production of potassium-rich lavas (Xu et al., 2019). The earliest strata in the Mangkang basin were deposited during the Cretaceous and are predominantly red in color and consist of silico-calcareous sandstones, calcareous siltstones, and mudstones, with gypsum interbeds (Zhang et al., 2005; Su et al., 2018). The Mangkang igneous rocks are mainly distributed in the Lawuxiang Formation, and the rock types are mainly trachyte, trachyandesite, monzonite porphyry, syenite-porphyry, and quartz monzonite porphyry (Xu et al., 2019).

We conducted paleomagnetic sampling at 24 sites (ME1–ME24) in the Mangkang area (29°53′26.34″N, 98°33′10.89″E) (Fig. 1), including sampling sites ME1–ME11 and ME18–ME24 obtained from trachyte intrusive mass, and sampling sites ME12–ME17 obtained from lower Cretaceous red beds that are intrusive contact with the trachyte rocks. Typically, more than 10 cores were sampled from each site, producing more than 240 independent paleomagnetic samples. The samples were drilled using a portable gasoline-powered drill system, and each sampling site was located using a high-precision handheld GPS, and the correction was made for the local magnetic field (International Association of Geomagnetism and Aeronomy, 2015). All oriented core

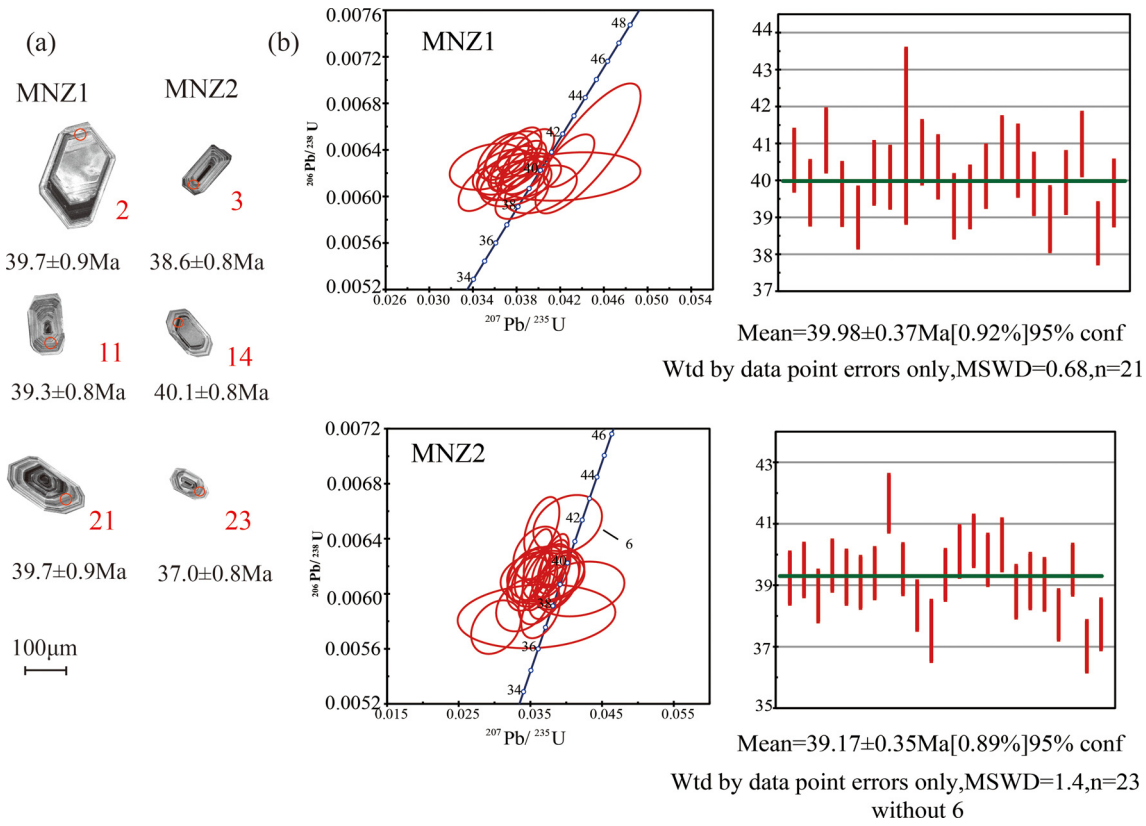


Fig. 2. (a) Cathodoluminescence images of representative zircons of samples MNZ1, MNZ2, and corresponding $^{206}Pb/^{238}U$ ages of the individual analyzed spots. (b) Concordia diagrams showing $^{206}Pb/^{238}U$ ratios in zircon grains in samples MNZ1, MNZ2, (left columns); a weighted average of apparent $^{206}Pb/^{238}U$ ages (right columns).

samples were cut into standard cylindrical specimens of 2.2 cm length and 2.54 cm diameter, and each core sample was processed into at least two standard specimens for rock magnetic and thermal demagnetization experiments. All paleomagnetic experiments were completed at the Key Laboratory of Paleomagnetism and Continental Reconstruction, Ministry of Natural Resources, Beijing.

3. U–Pb zircon dating

Two samples were selected for U-Pb dating: one sample from the bottom (MNZ1) and another from top (MNZ2) of the sampled section. Zircons were separated using standard density and magnetic separation methods in the Laboratory of Geophysical Exploration in Langfang, China. The zircon grains are euhedral to subhedral (100–300 μm long and 50–100 μm wide) with distinct oscillatory zonation in most grains. The Th/U ratios of the zircons are concentrated in the range of 0.4–0.8, suggesting that the zircon grains are of magmatic origin.

Zircon U-Pb ages data were obtained using the SHRIMP II instrument at the National Science and Technology Infrastructure Platform, Beijing Ion Probe Center (Institute of Geology, Chinese Academy of Geological Sciences). The results for these two samples show a highly concentrated grouping of data. As shown in Fig. 2, the weighted average data for the sample from the bottom of the sampling section (sample MNZ1) is 39.98 ± 0.37 Ma (2σ, MSWD=0.68, n=21), and the weighted average data of the sample from the top (sample MNZ2) is 39.17 ± 0.35 Ma (2σ, MSWD=1.4, n=23). Therefore, the estimated age of the trachyte intrusive mass in the sampling area at Mangkang lies within the range of 39.2 to 40 Ma, which is consistent with the age estimate of ~36–41 Ma obtained by previous studies in the Mangkang area (Su et al., 2018; Xu et al., 2019).

4. Rock magnetism

4.1. Rock magnetic measurements and results

A series of rock magnetic measurements were conducted to characterize the magnetic mineral assemblages and their contribution to the remanence of the upper Eocene trachyte rocks and Cretaceous sandstone red beds of the Mangkang area. Three representative samples were selected from trachyte intrusive mass (ME3-5 and ME24-6) and sandstone (ME13-4), based on the relative distribution of sampling sites and lithologic characteristics. These samples were used for isothermal remanent magnetization (IRM) component analysis (Kruiver et al., 2001), hysteresis loop measurements, and thermal demagnetization of three-component IRM (Lowrie, 1990). The IRM curves and hysteresis loops were measured using a Lakeshore 8600 vibrating sample magnetometer. The thermal demagnetization of three-component IRM was performed by applying magnetic direct current (DC) fields of 2.45 T, 0.4 T, and 0.12 T to sample Z, Y, and X axes, respectively, followed by systematic thermal demagnetization (Hirt and Lowrie, 1988; Lowrie, 1990). All these experiments were conducted at the Key Laboratory of Paleomagnetism and Tectonic Reconstruction, Ministry of Natural Resources, Institute of Geomechanics, Chinese Academy of Geological Sciences.

The results of the rock magnetic experiments are shown in Fig. 3. The IRM acquisition curves of sample ME3-5 were fitted with three components. Component 1, with $B_{1/2}$ (the applied DC field at which half of the saturation isothermal remanent magnetization [SIRM] is attained) of 14.8 mT (DP=0.08 mT), comprises 8.7% of the SIRM; component 2, with $B_{1/2}$ of 58.9 mT (DP=0.3 mT), constitutes 88% of the SIRM; and component 3, with $B_{1/2}$ of 631.0 mT (DP=0.34 mT), comprises 3.3% of the SIRM. We interpret components 2 and 3 as representing mag-

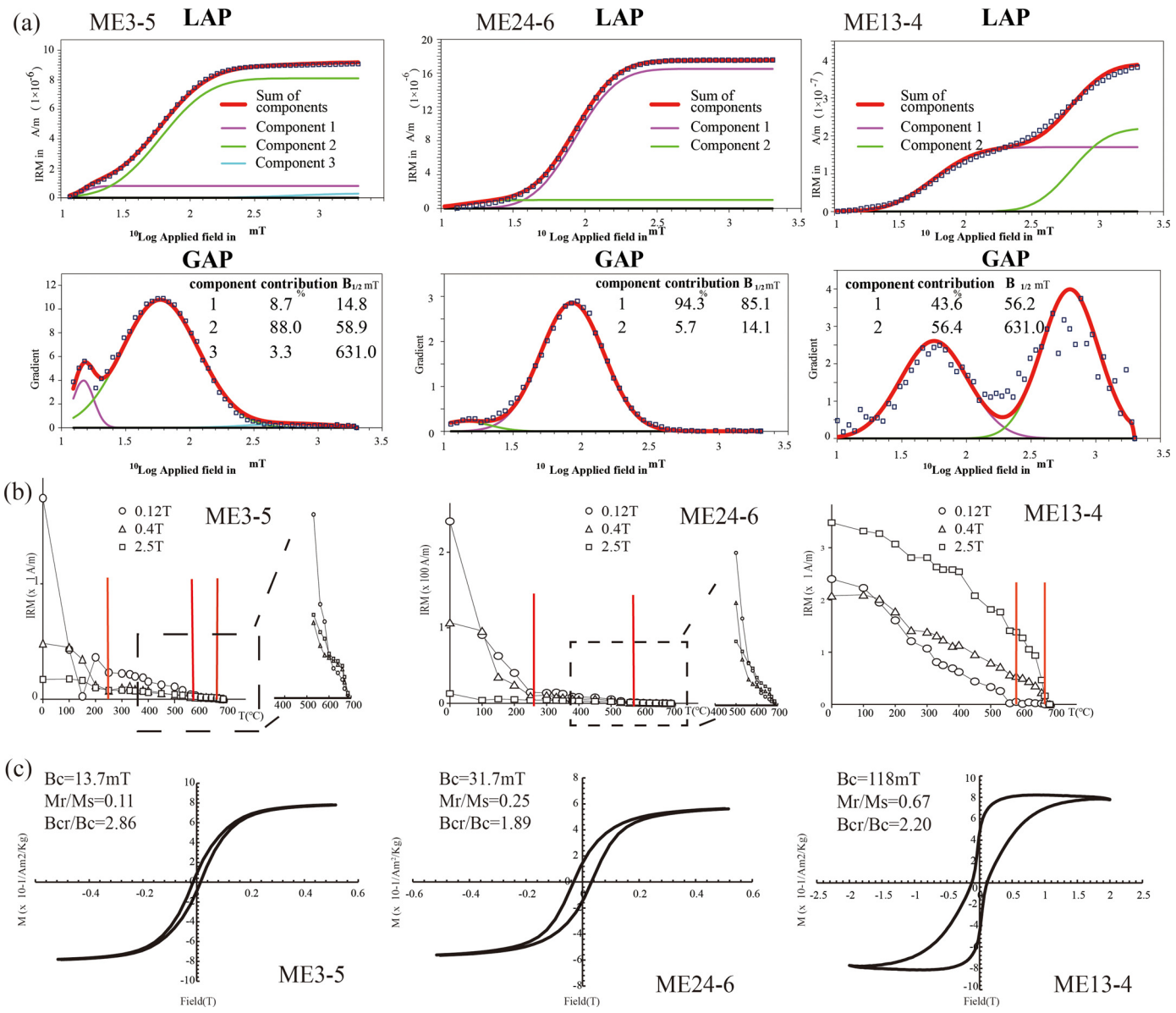


Fig. 3. (a) Result of IRM component analysis (Kruiver et al., 2001). Squares are measured data points. The components are marked with lines. LAP, linear acquisition plot; GAP, gradient acquisition plot; $B_{1/2}$, the field at which one half of the SIRM is reached; DP, the dispersion parameter, indicating the width of the distribution. (b) Results of thermal demagnetization of a three-component IRM, with direct current fields of 2.5 T, 0.4 T, and 0.12 T, applied successively to the Z, X, and Y axes, respectively (Lowrie, 1990). (c) Results of hysteresis loop measure.

netite and hematite, respectively, while component 1 may be fine-grained magnetite. For sample ME24-6, component 1, with $B_{1/2}$ of 85.1 mT (DP=0.1 mT), comprises 94% of the SIRM; and Component 2, with $B_{1/2}$ of 14.1 mT (DP=0.3 mT), comprises 6% of the SIRM. Components 1 and 2 are magnetite and fine-grained magnetite, respectively. Thermal demagnetization of the three-component IRM of the trachyte samples shows that the intensity of the soft component shows an inflection at ~250 °C and the intensities of the soft, medium and hard components show rapid decreases at ~500–560 °C and ~600–680 °C, indicating magnetite and hematite, respectively.

The IRM acquisition curves of representative red sandstone samples ME13-4 were fitted with two IRM components, Component 1, with $B_{1/2}$ of 56.2 mT (DP=0.1 mT), comprising 44% of the SIRM; and component 2, with $B_{1/2}$ of 631 mT (DP=0.3 mT), comprises 56% of the SIRM. Thermal demagnetization of the three-component IRMs showed that the soft, medium, and hard components exhibit inflections at ~580 °C, after which all three components are unblocked at ~680 °C, indicating the presence of magnetite and hematite.

The hysteresis loops of the trachyte samples are all wasp-waisted. Sample ME3-5 has B_c and B_{cr} values of 13.7 mT and 39 mT, respectively, and M_{rs}/M_s of 0.10. Sample ME24-6 has B_c and B_{cr} values of 31.7 mT and 60 mT, respectively, and M_{rs}/M_s of 0.25. These results indicate that the trachyte samples are dominated by low-coercivity magnetic minerals. The red sandstone sample is also wasp-waisted. Sample ME13-4 has B_c and B_{cr} values of 118 mT and 260 mT, respectively, and M_{rs}/M_s of 0.67, indicating the co-occurrence of magnetite and hematite.

The rock magnetic results indicate that the main magnetic carriers for the trachyte samples are magnetite and hematite. The fact that the thermal demagnetization of three-component IRM curves does not show clear unblocking temperatures of 580 °C and 680 °C likely indicates some degree of Ti substitution in the Fe oxides. In the red sandstone samples, hematite and magnetite are the dominant magnetic carriers.

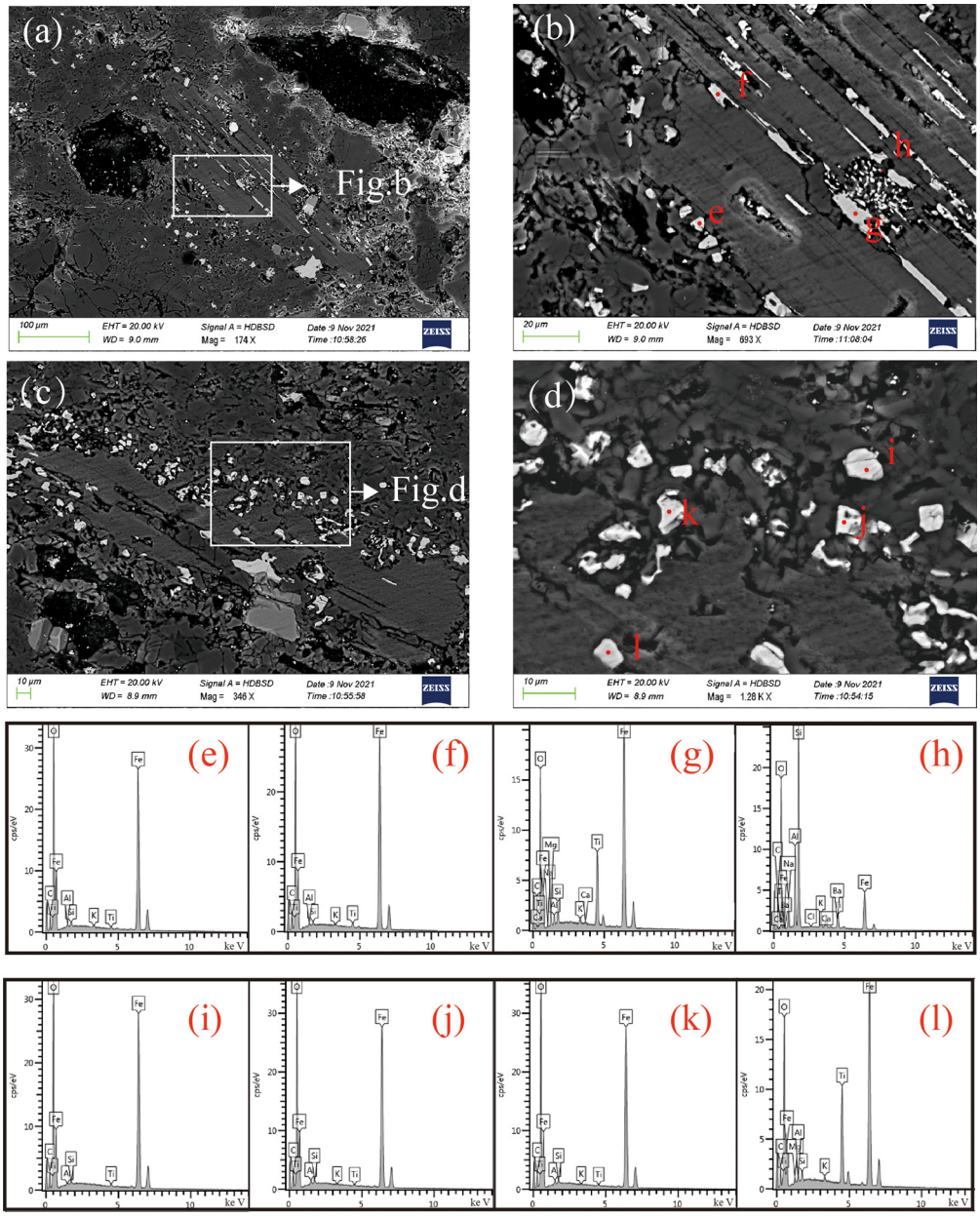


Fig. 4. SEM observations and EDS analyses of the representative specimen (a - d) Solid circles in the graphs indicate the points analyzed by EDS (e–l); the corresponding results of which are presented.

4.2. EDS analyses

After identifying the magnetic carriers in the volcanic samples from the Mangkang area, it is important to determine whether the natural remanence is of primary origin and whether the hematite carrier in the volcanic samples is the result of low-temperature oxidation (LTO) of titanium-rich magnetite. Therefore, we selected representative samples for field emission scanning electron microscopy (FESEM) and energy dispersive X-ray spectrometer (EDS) analysis. The results showed that the volcanic samples are rich in Fe oxide grains with irregular shapes and different sizes, ranging mainly from a few microns to hundreds of nanometers in diameter (Fig. 4). The EDS analyses indicated that these Fe oxides contain only a small amount of Ti, and the Backscattered electron (BSE) images show no evidence of an oxidation edge on the Fe oxide grains. These findings, combined with the rock magnetic results described above, demonstrate that both the magnetite and hematite were formed by initial crystallization, rather than by a long-term LTO process after cooling (Swanson-Hysell et al., 2011). This conclusion was confirmed by the NRM thermal demagnetization results (below) which show that both magnetite and hematite carry similar remanence directions. The Fe oxide grains have no obvious oxidized rims also indicates that no remagnetization has affected these rocks, also supporting the primary origin of the remanent magnetization.

5. Paleomagnetic results

The samples were subjected to stepwise thermal demagnetization using an ASC TD-48 thermal demagnetizer and the remanence was measured with a JR-6 spinner magnetometer or 2G-755 cryogenic magnetometer. The initial temperature interval was 50-100 °C, which was then decreased to 20-30 °C for temperatures from 300 °C to 680 °C. The demagnetization behavior of the specimens was analyzed using Zijderveld diagrams (Zijderveld, 1967), and then the characteristic remanent magnetization (ChRM) of all

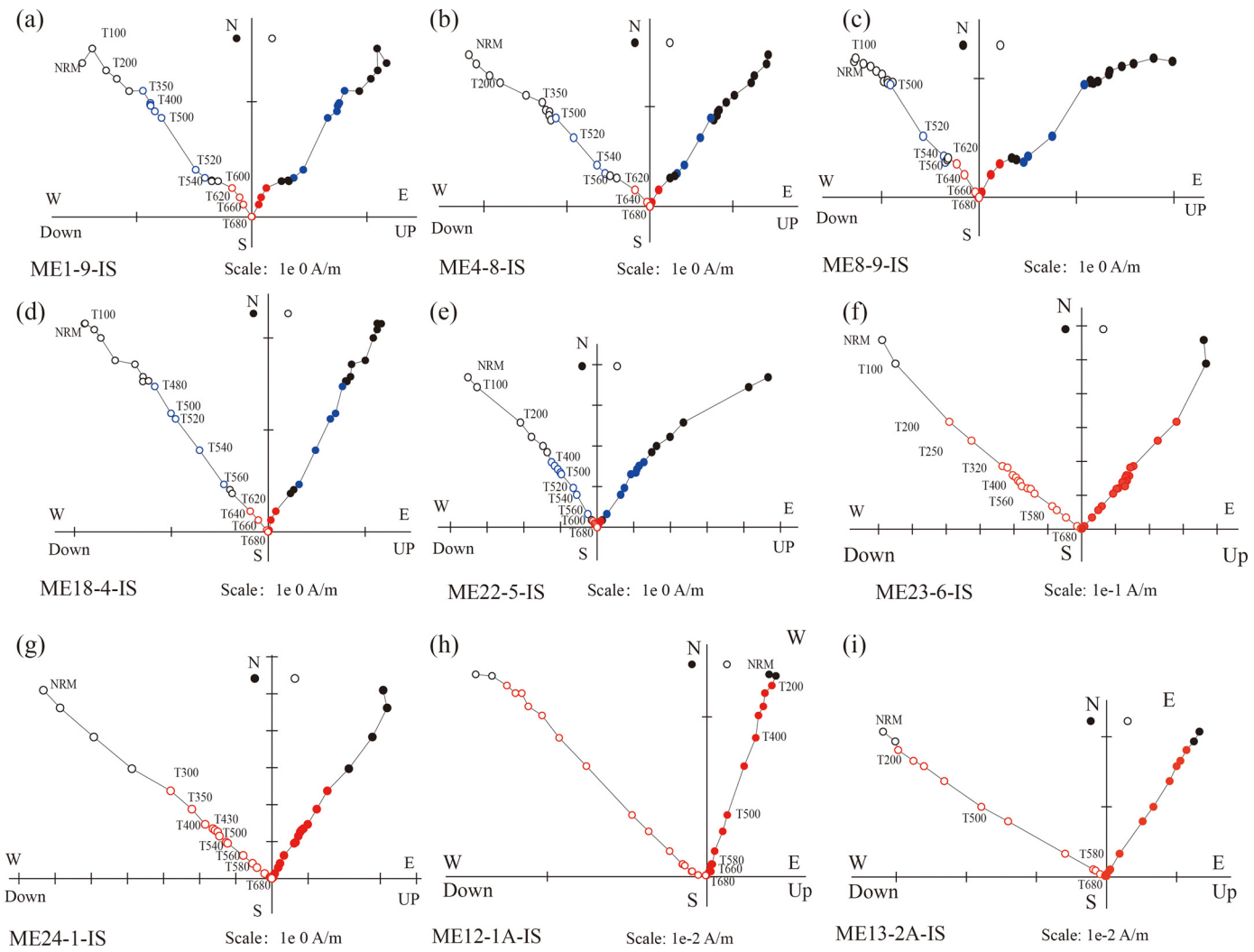


Fig. 5. Orthogonal projections of magnetization vector end-points for representative specimens of trachyte specimens and red sandstone specimens. High-laboratory unblocking temperature magnetization components were isolated using principal components analysis (Kirschvink, 1980). Solid/open symbols are the horizontal/vertical projections in geographic coordinates, the red symbol is for "high-laboratory unblocking temperature component" and the blue symbol is for "middle-laboratory unblocking temperature component". (a-f) are trachyte specimens, (g-i) are sandstone specimens.

the specimens was determined using principal component analysis (Kirschvink, 1980). The site-mean directions were calculated using Fisher statistics (Fisher, 1953).

The red sandstone samples exhibit stable behavior during progressive thermal demagnetization, and a single magnetization component decayed linearly towards the origin, up to 680 °C (Fig. 5). The direction of this component is D_g=21.8°, I_g=54.6°, k_g=36.1, α_{95}=3.4° before tilt correction, and D_s=38.0°, I_s=31.2°, k_s=27.0, α_{95}=4.0°, n=49 specimens after tilt correction (Fig. 6). The precision parameter (k) decreased (from 36.1 to 27) and α_{95} increased (from 3.4° to 4.0°) after tilt correction.

This result differs greatly from previous Early Cretaceous paleomagnetic data (D=46.8°, I=54.5°) for the Mangkang area (Tong et al., 2015, Huang and Opdyke, 1992), and therefore we infer that the Cretaceous red sandstones in contact with the igneous rocks in the area that we sampled were subsequently remagnetized by the intrusive igneous rocks during the late Eocene.

The trachyte samples exhibit two nearly parallel magnetization components during progressive thermal demagnetization, within the ranges of 300 °C to 580 °C (middle temperature component, MTC) and 600 °C to 680 °C (higher temperature component, HTC). The direction of the MTC is D=28.1°, I=35.7°, k =43.5, α_{95}=1.9°, n=132 specimens; and that of the HTC is D=26.1°, I=37.7°, k=42.9, α_{95}=1.8°, n =150 specimens.

Sampling sites 23 and 24 are close to the red sandstone sites. In comparison to the other sampling sites with two magnetization components, most of the samples from these two sampling sites have a single magnetization component (Fig. 5), with D=30.5°, I=46.8°, k=132.1, α_{95}=3.2°, n=16 specimens. Coincidently, the magnetization direction from the marginal zone of the volcanic rock is consistent with the 40% unfolding direction (D=30.7°, I=45.9°, k=33.5, α_{95}=3.6°) of the tilted red beds, confirming that the red beds were remagnetized by the igneous rocks while the red beds were tilted by ~60%. On the other hand, the HTC (D=28.5°, I=35.7°, k=68.3, α_{95}=2.0°, n=78 specimens) and MTC (D=31.9°, I=35.1°, k=63.2, α_{95}=2.0°, n=78 specimens) of sampling sites 1–9, distant from the red sandstone sites, are different from the secondary magnetization in the red beds. Statistically, the angular distance of the mean remanent directions obtained from sites 1–9 and the red sandstone is 19.4° ± 3.1°, compared to the directional difference (9.4° ± 3.6°) from those of sites 23 and 24 and the red sandstone; the angular distance of the remanent directions obtained from sites 1–9 and the 40% unfolding direction of red sandstone is 10.2° ± 3.2°, compared to the direction difference (0.9° ± 3.7°) from those of sites 23

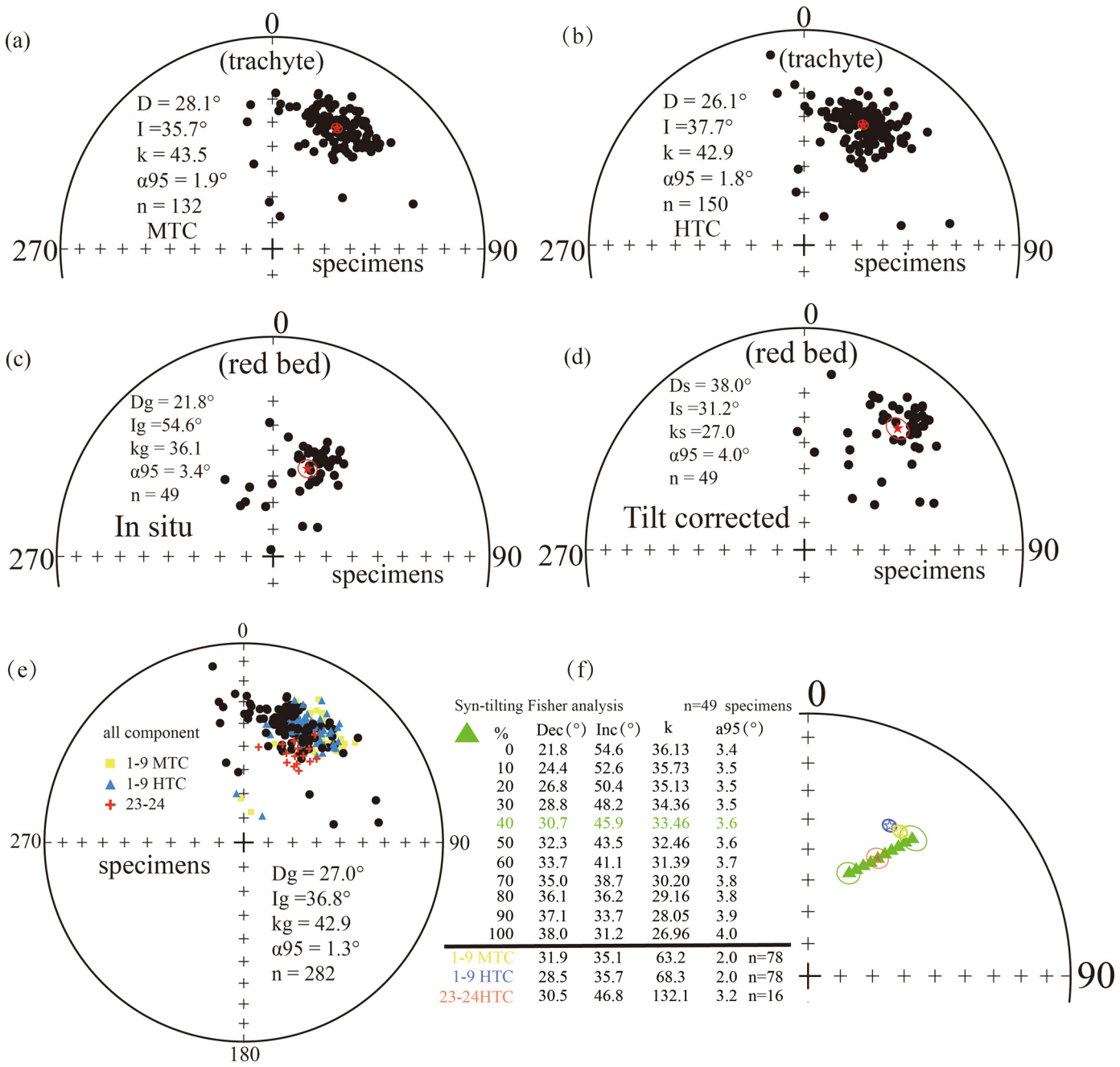


Fig. 6. (a-b) Equal-area projections of section mean directions of "middle-laboratory unblocking temperature component" or "high-laboratory unblocking temperature component"; (c-d) Equal-area projections of section mean directions of sandstones before and after tilt correction; (e) Equal-area projections of section mean directions of "All components"; (f) Syn-tilting Fisher analysis of results from sandstones.

and 24 and 40% unfolding direction of the red sandstones. The red sandstone is interpreted to have been remagnetized during heating by the intrusive mass; therefore, the closer to the red sandstone sites, the greater the similarity of the magnetization of the intrusive mass and red sandstone. In addition, due to the rapid cooling of the marginal zone of intrusive mass, the magnetic remanence carried by magnetite and hematite recorded almost the same geomagnetic field direction and exhibit a single magnetization component, yet the samples within the intrusive mass exhibited two components (HTC and MTC), possibly due to the slower cooling and different unblocking temperatures of magnetite and hematite. Because the cooling time may have been short, the two magnetizations acquired are similar. Another possibility is that the oxidation of magnetite to hematite is in the late stage of the cooling process, so that the HTC and MTC directions are very similar. Regardless of which of these two possibilities is correct, these two components may provide a geomagnetic record over a relatively long interval, rather than an instantaneous record of the field. Therefore, we opted to average the HTC and MTC magnetizations, rather than use a single high-temperature component. Combining these two components yields the mean geomagnetic field direction of D=27.0°, I=36.8°, k=42.9, α_{95}=1.3°, n=282 specimens. Table 1 lists the mean directions for the MTC, and HTC, together with the combined directions of the two components, and there is no statistically significant difference between them.

The instantaneous magnetic field can be recorded during the rapid cooling of magma which thus provides a spot reading of the

Table 1
Characteristic remanent magnetization obtained from the Late Eocene samples of the Mangkang.

Components	n	Rock Type	Observed			paleopole			paleosecular	variation test	Pass (Y/N)
			Dec. (°)	Inc. (°)	α_{95} (°)	Plat. (°)	Plon. (°)	A_{95} (°)	A_{95}max (°)	A_{95}min (°)	
1-9MTC	78	trachyte	31.9	35.1	2.0	-	-	-	-	-	-
1-9HTC	78	trachyte	28.5	35.7	2.0	-	-	-	-	-	-
23-24HTC	16	trachyte	30.5	46.8	3.2	-	-	-	-	-	-
Red bed	49	sandstone	38.0	31.2	4.0	-	-	-	-	-	-
MTC	132	trachyte	28.1	35.7	1.9	62.7	203.7	2.0	3.80	1.71	Y
HTC	150	trachyte	26.1	37.7	1.8	64.9	202.8	1.9	3.46	1.61	Y
ALL	282	trachyte	27.0	36.8	1.3	63.9	203.3	1.4	2.33	1.25	Y

n — the number of specimens used for Fisher statistic; Dec — Declination; Inc — Inclination; α_{95} — the radius of cone at 95 percent confidence level of sample-mean direction; Plat. — paleolatitude; Plon. — paleolongitude; A_{95} — the radius of cone at 95 percent confidence level of paleopole; Y/N — yes/no.

geomagnetic field (Cox, 1970; McFadden et al., 1991). The effect of paleosecular variation of the Earth's magnetic field on the magnetization of a sequence of volcanic rocks needs to be assessed. The geochronologic data indicate that the age difference between the top and bottom of the sampling section is ~ 0.81 Ma ± 0.02 Ma, which is sufficiently long to average out the paleosecular variation of the Earth's magnetic field. In addition, the A_{95} envelope method of Deenen et al. (2014) was used to assess the reliability of the data: $A_{95}max=82\times N^{-0.63}$, $A_{95}min=12\times N^{-0.40}$, where N equals the number of specimens. The A_{95} of a reliable dataset should be within the range of $A_{95}max=2.33°$, $A_{95}min=1.25°$. The site-mean paleopole for 282 specimens using Fisher statistics is located at (63.9°N, 203.3°E) with $A_{95}=1.4°$. The average VGP of all specimens ($A_{95}vgp=1.4°$) is within the upper and lower limits of the A_{95} envelope ($A_{95}max=2.33°$, $A_{95}min=1.25°$). To confirm that our data are not significantly affected by paleosecular variation, we also performed the paleosecular variation test for the magnetite and hematite components, individually. The A_{95} of the mean magnetite component is 2.0°, n=132 specimens ($A_{95}max=3.80°$, $A_{95}min=1.71°$), and the A_{95} of the mean hematite component is 1.9°, n=150 specimens ($A_{95}max=3.46°$, $A_{95}min=1.61°$). Thus, both datasets passed the paleosecular variation test (Table 1).

6. Discussion

Igneous rocks often preserve accurate records of the ambient geomagnetic field, because unlike sedimentary rocks they are not affected by compaction-induced inclination shallowing (Tauxe and Kent, 2004; Tan et al., 2003). The paleomagnetic data obtained in this study have averaged out the paleosecular variation, and in addition, SEM BSE images show that the edges of the Fe oxide grains lack evidence of oxidation, indicating that the sampled section was not affected by orogenic fluids. Thus, the ChRM of the trachyte samples provides a primary record of the geomagnetic field.

6.1. Latitudinal shortening of the southeastern Tibetan Plateau since the late Eocene

Ding and Zhong (2013) noted that the width of the central part of the Lhasa terrane exceeds 300 km, but that crossing the EHS from west to east, the width of Lhasa terrane in the Chayu area is reduced to ~200 km, and the width of the Tengchong terrane on the southeastern side of the EHS is only 90 km. They also reported that the eastward extent of the Yarlung Zangbo suture zone (YZS) between India and Asia extends to the Tuting and Tidding areas in the eastern EHS. The Gangdese Arc of the southern part of the Lhasa terrane correlates with the Chayu Arc in the eastern EHS. However, the rock-stratigraphic unit of Tethys-Himalayan terrane in northern India exhibits a large difference across the EHS: the Greater Himalaya crystalline basement and Tethys-Himalayan sedimentary rocks are exposed in the Milin area of the western EHS, and Tethys-Himalayan sedimentary rocks are absent in the southern parts of the Tuting and Tidding areas in the eastern EHS (Ding and Zhong, 2013) (Fig. 7). These geologic investigations suggest that strong tectonic deformation around the EHS occurred due to northward compression in the central Tibetan Plateau, to the west of the EHS, causing the extrusion of Greater Himalayan crystalline basement in the Namche-Barwa Syntaxis, and deformation induced by NNW-SSE strike-slip faults in the eastern EHS during the India-Eurasia collision since the late Eocene. The Lhasa and Qiangtang terranes were shortened eastward around the EHS and were then rotated towards the San-Jiang transverse mountains. All these observations indicate changes in the tectonic regime in and around the EHS.

Mangkang is in the southeastern part of the Qiangtang terrane. In the region from the Bangong-Nujiang suture zone to the Lhasa terrane, the Gangdese Arc in the south was underthrusted by the extension of the Greater Himalayan crystalline rock basement that formed the EHS. We used the estimated paleoposition of the Mangkang area to assess the crustal shortening between south Qiangtang and stable Eurasia, between the EHS and South Qiangtang, and between the EHS and stable India, following the India-Eurasia collision since the late Eocene. Our paleomagnetic results for the Mangkang area are interpreted to suggest that its paleolatitude was 20.5° ± 0.9°N; thus, latitudinal shortening of 15.4° ± 2.1° (1694 ± 231 km) took place north of the Mangkang area with respect to Eurasia since ~40 Ma, based on a comparison with the 40 Ma Eurasian reference pole (Torsvik et al., 2012). The magnitude of crustal shortening between Qiangtang and stable Eurasia is consistent with the 13.0° ± 6.6° (1435 ± 726 km) of latitudinal difference obtained from volcanic rocks of ages between 38.6 ± 0.5 Ma and 35.4 ± 2.4 Ma in the interior of the Qiangtang terrane (Lippert et al., 2011), and also with the 11.2° ± 2.9° (1230 ± 319 km) latitudinal difference based on Eocene paleomagnetic results from the Gongjue basin in the eastern part of the Qiangtang terrane during the 56–43 Ma time interval (Tong et al., 2017) (Fig. 8) (Table 2).

The crustal shortening between the EHS and stable India is ~5.8° ± 2.3°, based on a comparison of the latitudinal difference of Mangkang and that inferred from the reference pole of India at 40 Ma (Torsvik et al., 2012) (Fig. 8). This is consistent with a series of balanced cross-section studies conducted in the Tethys-Himalaya terrane, which indicated that a total of 600–740 km of latitudinal crustal shortening has occurred within the Tethys-Himalaya terrane (Parsons et al., 2020). This suggests that ~64–80% crustal shortening has occurred in the western segment of the Tethys-Himalaya terrane, and ~75–77% crustal shortening in the central segment, with the greatest (82%) crustal shortening occurring in the eastern segment of the Tethys-Himalaya terrane in the EHS (Ningthoujam et al., 2015; Long and McQuarrie, 2010; Parsons et al., 2020).

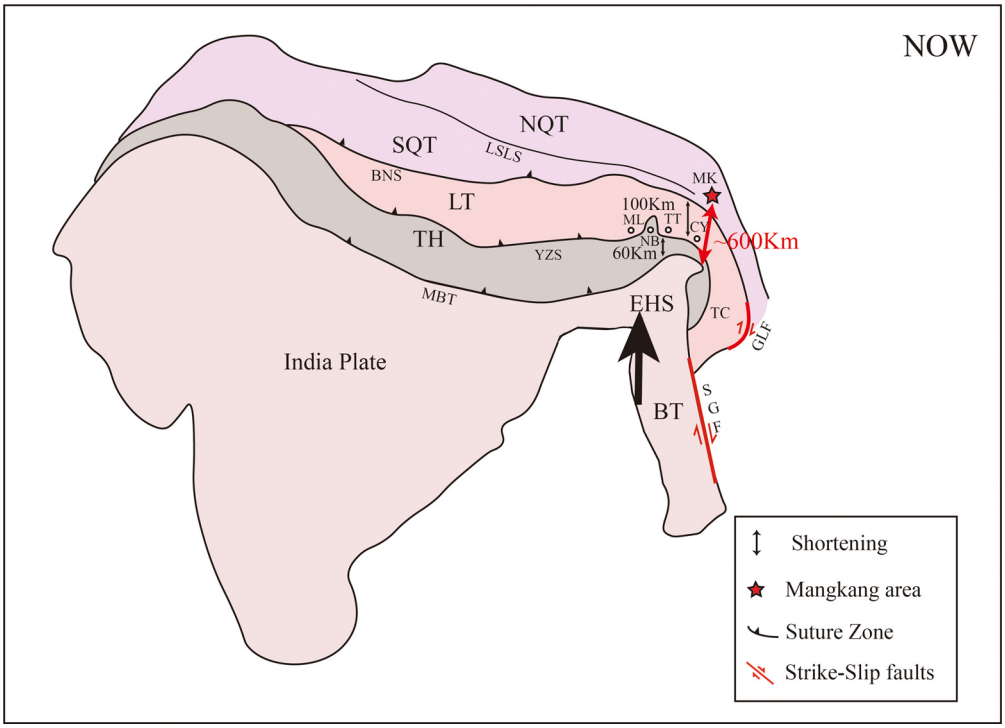


Fig. 7. Schematic map of Tibet, and the location of intracontinental shortening. SQT, South Qiangtang terrane; NQT, North Qiangtang terrane; LT, Lhasa terrane; BT, Burma block; CY, Chayu area; TC, Tengchong block; MK, Mangkang area; TH, Tethyan Himalaya; EHS, Eastern Himalaya Syntaxis; ML, Milin area; TT, Tuting and Tidding areas; BNZ, Bangong-Nujiang suture zone; NB, Namche Barwa Syntaxis; GLF, Gaoligong strike-slip fault; SGF, Sagaing strike-slip fault; YZS, Yarlung Zangbo suture zone; LSLS, Longmu Co-Shuanghu-Langcang River suture zone; MBT, Main boundary thrust.

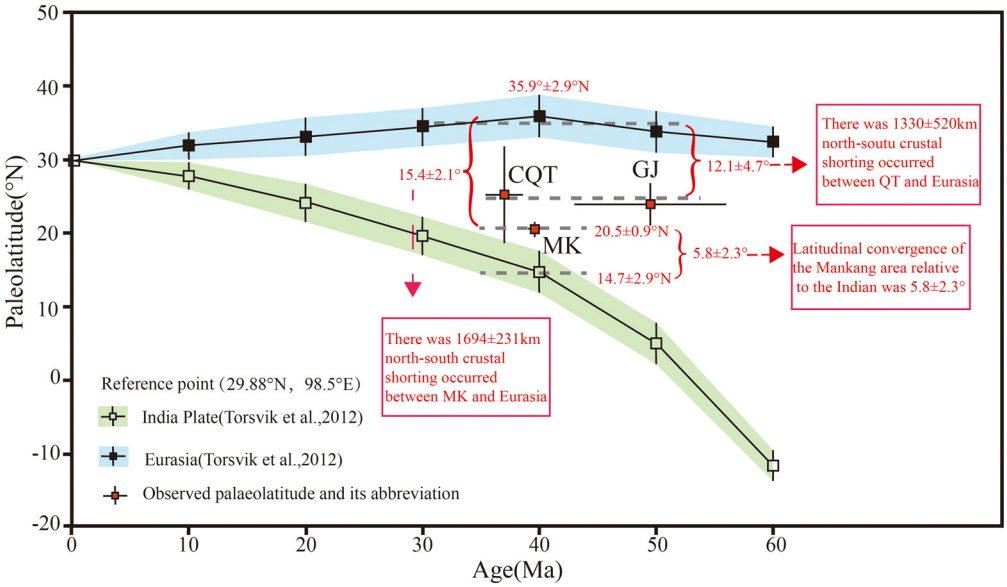


Fig. 8. Paleolatitude plots for Eurasia, the Indian craton, and the Qiangtang terrane. The shaded areas and vertical bars display the errors of the paleolatitudes. The expected and observed paleolatitudes were calculated for the reference point at 29.88°N, 98.5°E.

6.2. Extrusion tectonics of the southeastern Tibetan Plateau since the Oligocene

Along with large-scale tectonic deformation in the northern EHS, our results also indicate a clockwise rotation of 19.1° ± 2.6° with respect to Eurasia for the Mangkang area since about 40 Ma. This estimate is consistent with estimates of clockwise rotation of 24.2° ± 3.3° (Tong et al., 2017) and 17.7° ± 3.3° with respect to Eurasia after 43 Ma (Zhang et al., 2018), in the Gonjo basin on the north side of the Mangkang area. Additionally, Li et al. (2020) concluded that the Gonjo basin has experienced ~30° of clockwise rotation since at least 41 Ma. A paleomagnetic study in the Nangqian basin indicated ~24° clockwise rotation during 41–35 Ma (Zhang et al., 2020). These results indicate that the areas on the front of the EHS consistently experienced 20–30° of clockwise rotation since ~40 Ma, suggesting that the southeastern part of the Tibetan Plateau was involved in a crustal-scale clockwise rotation system, in the course of the formation and tectonic evolution of the EHS.

Previous paleomagnetic studies of the Cretaceous and Cenozoic strata of the EHS in the southeastern part of the Tibetan

Table 2
Cenozoic paleomagnetic data obtained from the Cenozoic strata, in the Qiangtang terrane.

Site (°N/°E)	Time (Ma)	N/n	Observed			paleopole			Paleolat	Translation	Rock Type	Reference
			Dec. (°)	Inc. (°)	α_{95} (°)	Plat. (°)	Plon. (°)	A_{95} (°)	(°)			
Qiangtang	Terrane											
32.7/85.6	35.4±2.4	9/-	15.5	40.0	7.2	73.1	207.6	8.8	22.8±5.2	14.2±6.8	Volcanic	Lippert et al., 2011
34.5/90.2	38.6±0.5	4/-	355.8	46.1	3.4	82.1	298.4	7.8	27.5±7.8	11.9±6.1	Volcanic	Lippert et al., 2011
31.0/98.2	56-43	43/-	35.5	41.6	3.2	57.9	192.1	2.9	23.9±2.9	11.2±2.9	Sandstone	Tong et al., 2017
29.9/98.5	39.6±0.4	-/282	27.0	36.8	1.3	63.9	203.3	1.4	20.5±0.9	15.4±2.1	Volcanic	This study
Eurasia	APWP		Reference Position (29.9°N, 98.5°E)									
	0					88.5	173.9	1.9	29.9±1.9			Torsvik et al., 2012
	10					86.7	150	1.8	31.9±1.8			Torsvik et al., 2012
	20					84.4	152.1	2.6	33.1±2.6			Torsvik et al., 2012
	30					83.1	146.5	2.6	34.4±2.6			Torsvik et al., 2012
	40					81.1	144.3	2.9	35.9±2.9			Torsvik et al., 2012
	50					78.9	164.7	2.8	33.8±2.8			Torsvik et al., 2012
	60					78.2	172.6	2.1	32.4±2.1			Torsvik et al., 2012
India	APWP		Reference Position (29.9°N, 98.5°E)									
	0					88.5	173.9	1.9	29.9±1.9			Torsvik et al., 2012
	10					87.2	240.4	1.8	27.7±1.8			Torsvik et al., 2012
	20					83.7	254.7	2.6	24.1±2.6			Torsvik et al., 2012
	30					79.7	281.7	2.6	19.6±2.6			Torsvik et al., 2012
	40					74.7	286.8	2.9	14.7±2.9			Torsvik et al., 2012
	50					65.1	278.4	2.8	5.0±2.8			Torsvik et al., 2012
	60					48.5	280.8	2.1	−11.6±2.1			Torsvik et al., 2012

N/n — the number of sampling sites/specimens used for Fisher statistic; Dec — Declination; Inc — Inclination; α_{95} — the radius of cone at 95 percent confidence level of sample-mean direction; Plat. — paleolatitude; Plon. — paleolongitude; APWP — apparent polar wander paths; A_{95} – the radius of cone at 95 percent confidence level of paleopole.

Plateau and its margins indicate major clockwise rotations since the late Oligocene. For example, from north to south, the accumulated clockwise rotation of the Lanping area in northwestern Indochina was up to 82.3° ± 9.3° since the late Eocene (Sato et al., 2001), and in the northwest part of the Shan Thai block, the clockwise rotation progressively decreased from 72.4° ± 14.3° in the west to 65.2° ± 9.2° in the east, since at least the Paleocene (Tong et al., 2016; Li et al., 2018). In addition, the accumulated clockwise rotation of the Pian Ma area on the eastern edge of the Tengchong block was up to 87.2° ± 11.8° since 40 Ma (Kornfeld et al., 2014). Tong et al. (2021) and Sun et al. (2022) both suggested that the southern edge of the EHS experienced major clockwise rotation with an average rotational rate of 3°/Myr since the late Oligocene and that the crustal clockwise rotation of the southeastern edge of the Tibetan Plateau commenced no earlier than ~25.7 ± 2.5 Ma, because of the continuous northward penetration of the Indian plate into the eastern segment of the Himalaya orogenic belt.

Compression-induced crustal tectonic deformation and thickening are one of the main consequences of the post-collision crustal shortening in southern Eurasia. Several paleo-elevation studies on the Tibetan Plateau suggested that crustal uplift had largely ended by 40–35 Ma in southern Tibet. For example, Xiong et al. (2020) suggested that the elevation of the Gongjue basin in the Qiangtang terrane increased rapidly from ~0.7 km to ~3.9 km during 56–40 Ma, based on the carbonate accumulation isotope paleotemperature method applied to carbonate-rich paleosols and marl. Rapid surface thickening occurred before 40–35 Ma, implying that latitudinal crustal shortening of the southeastern Tibetan Plateau mainly occurred during this interval; subsequently, the southeastern Tibetan Plateau likely remained stationary and did not experience significant crustal shortening. Tong et al. (2017) proposed that the South Qiangtang and Lhasa terranes experienced crustal shortening during 53–43 Ma, after which the crustal shortening was transferred to the northern side of the South Qiangtang terrane, since at least ~43 Ma. This evidence suggests that the shortening due to the India-Asia collision had almost ended, and the estimated 5.8° (~600 km) of shortening we obtained for Mangkang relative to the northeast corner of India represents the northward wedging of the northeast corner of India. The northward wedging process of the northeast corner of India also led to greater intra-continental shortening in the Mangkang area, relative to Eurasia, as also occurred in the central part of the Qiangtang terrane.

Recent studies on the southeastern margin of the Tibetan Plateau and the West Burma block have elucidated the process of formation of the EHS in the late Eocene. Westerweel et al. (2019) suggested that the West Burma block rotated from east-west to north-south spreading during 90–40 Ma. Since ~29 Ma, the Indian plate began to be subducted obliquely beneath the West Burma block, represented by the rapid uplift and cooling of volcanic arcs in the northern West Burma block (R. Y. Li et al., 2013), which caused the northward movement of the West Burma block. Subsequently, during 28–26 Ma, the West Burma block and the Indian plate began to be shortened northward into the Eastern Himalayan orogenic belt to form the EHS, further compressing the foreland (Westerweel et al., 2020; Tong et al., 2021), and initiating the right-lateral strike-slip of the SGF (Fig. 9), which was coincident with the northward movement of the EHS by ~600 km since the Oligocene.

The 5.8° (~600 km) of shortening included shortening within both the Lhasa terrane and the Himalayan orogen. The shortening within the Lhasa terrane was estimated to be ~100 km, caused by the reduction in the width of the Lhasa terrane itself, in the vicinity of the EHS and between the EHS and Mangkang in the BNZ. The results of balanced cross-section studies (Ningthoujam et al., 2015; Long and McQuarrie, 2010; Parsons et al., 2020) indicate that the shortening of the eastern side of the Himalayan orogen (82%) was greater than in the central part of the orogen (75–77%), and therefore the eastern side is ~60 km shorter (82% minus 75%) than the central part (Fig. 7). Because the time duration of the shortening of these two terranes is unclear, the northward displacement of the West Burma block is estimated to be 400-600 km. If the shortening of the Lhasa terrane (by 100 km), and the shortening of the Himalayan orogen (by 60 km) occurred after the northward indentation of the Burma block into Eurasia, then this 600 km of latitudinal shortening represents the northward displacement of

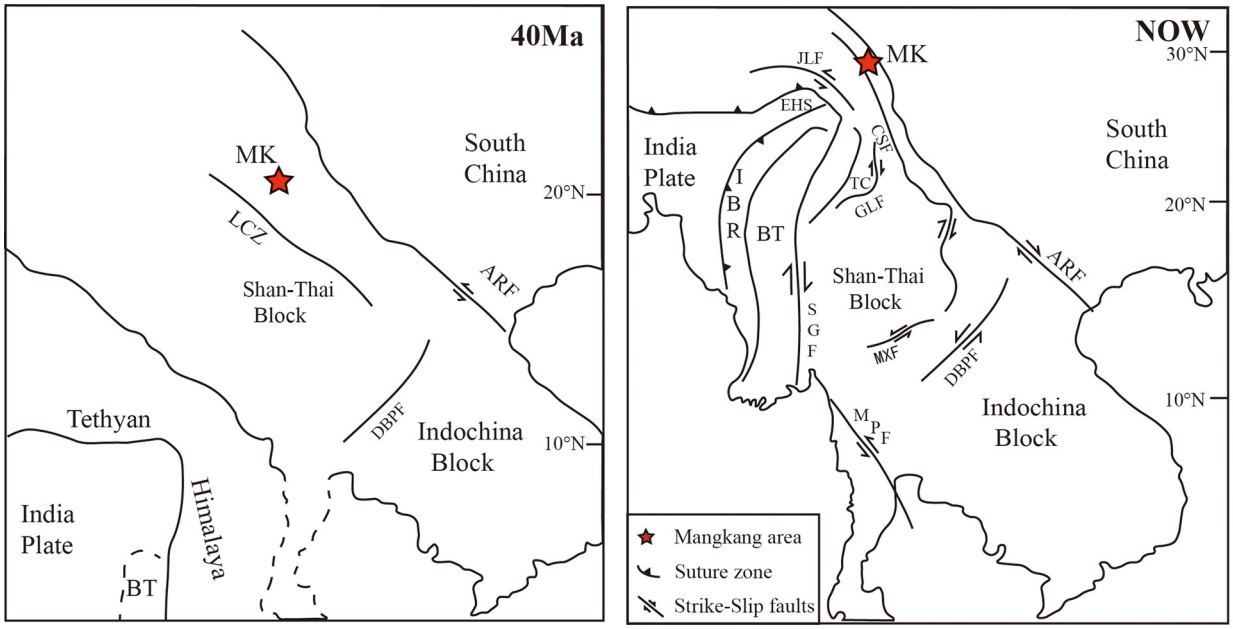


Fig. 9. Cenozoic tectonic evolution of the southeastern edge of the Tibetan Plateau illustrated based on the two important time nodes. ARF, Ailao Shan-Red River strike-slip fault; CSF, Chongshan strike-slip fault; DBPF, Dien Ben Phu strike-slip fault; GLF, Gaoligong strike-slip fault. JLF, Jiali strike-slip fault; SGF, Sagaing strike-slip fault; MPF, Mae Ping strike-slip fault; MXF, Mengxing strike-slip fault; LCR, Langcang River suture zone; MK, Mangkang area; IBR, Indo-Burma Ranges; BT, Burma terrane; TC, Tengchong block.

the Burma block. Moreover, if this 160 km of shortening occurred before the indentation of the Burma block into Eurasia, then the northward displacement of the Burma block should be ~440 km (600 km minus 160 km).

The Indochina Block is generally thought to have been extruded laterally since the late Eocene, due to the continuous collision of India and Asia (Tapponnier et al., 1982). The Lancang fault between the Shan-Thai block and the Indochina block shows an insignificant amount of strike-slip displacement, even if it is generally suggested that the Shan-Thai block and the Indochina block were extruded. However, paleomagnetic studies have failed to detect any significant southward displacement of the Shan-Thai block, based on inclination-corrected red bed data, the intrusive volcanic mass data in the Gaoligong fault, and the Cenozoic secondary remanences data (Kornfeld et al., 2014; Tong et al., 2016, 2021).

The western boundary of the Shan-Tai block is the SGF, which is adjacent to the West Burma block. Morley (2014), Morley and Arboit (2019) suggested that the SGF was displaced by 400–600 km, based on the offset of the ophiolite belt during the middle Oligocene (28–27 Ma). This amount of strike-slip offset is consistent with the wedging of the West Burma block into Eurasia. This indicates that the northward displacement of the West Burma block absorbed the strike-slip of the SGF, while the Shan-Thai block underwent no significant southward displacement accompanied by major clockwise rotation. This inference is consistent with Cenozoic paleomagnetic results from the Shan-Thai block. Although the Shan-Thai block may have undergone a kinematic reversal, geologic evidence suggests that the magnitude of this reversal was small. For example, the Nantinghe fault, Menglian fault, Jinghong fault, and a series of NE-striking faults, only indicate a kinematic reversal (northward displacement) of less than 50 km (Wang et al., 2022). Therefore, we conclude that the Shan-Thai block underwent an insignificant degree of southward displacement.

7. Conclusions

1. The paleomagnetic direction obtained from 282 specimens of upper Eocene (~40 Ma) trachyte intrusive mass rocks in the Mangkang area is D=27.0°, I=36.8°, k=42.8, α_{95}=1.3°, n=282 specimens. This result indicates that the Mangkang area was located at the paleolatitude of 20.5 ± 0.9° N during the late Eocene and was rotated clockwise 19.1 ± 2.6° with respect to stable Eurasia.

2. The Mangkang area has been continually pushed northward by the EHS since the late Eocene, resulting in a clockwise rotation and additional intra-continental shortening. Based on the paleolatitude of the Mangkang area, the northeast corner of the Indian plate and the West Burma block have continuously moved northward by ~600 km since 40 Ma.

3. The northward wedging of the EHS (~600 km) implies that the ~400–600 km of right-lateral slip in the Sagaing fault zone was absorbed by the northward wedging of the West Burma block, which implies a minor southward shift of the northern Shan-Thai block.

CRediT authorship contribution statement

Zijian Zhang: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft. **Yabo Tong:** Conceptualization, Funding acquisition, Investigation, Methodology, Writing – review & editing. **Shuchen Jin:** Formal analysis, Investigation. **Xinxin Sun:** Formal analysis, Investigation. **Lifu Hou:** Formal analysis, Investigation. **Junling Pei:** Investigation, Methodology. **Zhenyu Yang:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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